

View Reviews

Paper ID

1204

Paper Title

VisionFood QAI: A Deep Learning Based Quality Inspection System for Food and Beverage Products

Track Name

Track-4: Data Science, Analytics, and Intelligent Decision Systems

Reviewer #2

Questions**1. Overview and strengths of the paper.**

VisionFood QAI goes after food and beverage defect inspection, and the literature survey is honestly one of the better parts of the paper. Sixteen or seventeen studies (more on that discrepancy below), grouped into four themes: CNN/transfer learning, YOLO detection, IoT/sensor fusion, and specialized methods like autoencoders and incremental learning. The gap analysis holds up too. Beverages really are underrepresented in the food QC literature, and I haven't seen another paper that tries to combine localization, explainability, and remediation guidance into one workflow. The per-study limitations tables (1 through 4) are specific rather than the usual vague "future work needed" filler, and the comparison against Liu et al. (the closest prior work) is fair and doesn't oversell the gap.

On the modeling side, three components are actually built and tested, not just sketched out on a whiteboard: a YOLOv11s cap-defect detector (mAP@0.5 of 0.995, recall 0.973), a MobileNetV3-Small classifier for cap quality using Focal Loss (86.5% accuracy), and a YOLOv8 water-surface detector feeding into a closed-form fill-ratio calculation. The two-pass zoom trick for catching small caps that get missed on the first detection pass is a smart, well-justified piece of engineering, not just architecture-shopping. I also want to give the authors credit for something a lot of papers skip: when the cap classifier missed its own accuracy target, they said so plainly instead of quietly moving the goalposts. That's the right instinct, and it should be encouraged, not punished in review.

2. Shortcomings in the paper (in detail to be addressed).

1. The title and abstract sell "VisionFood QAI" as a full system, SKU routing, a REMEDY severity engine, XAI visualizations, a full-stack dashboard with FastAPI, React, and MongoDB. None of that has a single evaluation number attached to it: no accuracy, no latency, not even a screenshot. Everything past Section VI is written in

“the system generates” / “the dashboard displays” language, future tense dressed up as present tense. Given this is labeled Capstone Phase 1, I get why the scope outran the build, but the abstract shouldn’t read as if the whole thing is done when what’s actually tested is three CV submodules.

2. The paper sets its own bar in Section II.A: pass/fail accuracy of 93% or better, latency under 100ms on GPU. Neither number shows up anywhere in Section VIII.

These are the two headline metrics the intro promises, and they’re just absent from the results.

3. Table 13 is cherry-picked, whether or not that was intentional. Table 9 shows the cap detector’s precision at 0.869 against a target of 0.93, a real miss, but precision doesn’t appear in Table 13 for that module at all. Same pattern with the water-surface detector: Table 12 has precision, recall, and F1 all under target, and Table 13 keeps only the two metrics that pass. Put every number in the summary table, not just the flattering ones.

4. The YOLO detector and the MobileNetV3 classifier are predicting the same three cap classes, and the paper claims the combined pipeline beats either model alone. Still, there’s no fusion logic described (what wins when they disagree?) and no evaluation of the combined output. Right now that’s an assertion, not a result.

5. Every metric reported is a validation-set number, and for two of the three modules it’s explicitly the best checkpoint on that same validation set. There’s no held-out test set anywhere in the paper. Picking your best epoch on validation and then quoting that epoch’s validation score as your final number is a well-known way to flatter your own results, even unintentionally.

6. The architecture claims to handle four packaging types (bottles, cans, wrappers, boxes), but only the bottle case (cap plus fill level) ever gets implemented or tested. The other three are architecture-only. Worth saying so directly in the text.

7. Small thing, but reference [18] (“Implicit bias of linear equivariant networks,” ICML 2022) is never cited in the body and has nothing to do with food inspection or the model families used here. Looks like a leftover from a bibliography template. Pull it or explain it.

8. The abstract says “16 recent studies,” but Tables 1 through 4 add up to 17 distinct references ([1] through [17]). Minor, but easy to fix.

9. Table 5 justifies the mAP@0.5 target of 88% as “competitive with Liu et al.,” but the paper only reports precision/recall/F1 for Liu et al., never an mAP figure. So the target is anchored to a number that isn’t actually in the comparison paper.

10. There’s no description anywhere of how big the custom cap or water-surface datasets are, class balance, or annotation process. Given class imbalance is explicitly cited as a reason for using Focal Loss, I’d want to know how many defective-cap images are actually in that validation set before trusting the 86.5% number much.

3. Overall evaluation

Borderline

4. Reviewer's confidence on the overall evaluation of the paper

High

Reviewer #3

Questions**1. Overview and strengths of the paper.**

The paper presents VisionFood QAI, a computer-vision-based inspection system for food and beverage packaging. The proposed architecture combines YOLOv11s for bottle and cap-defect detection, MobileNetV3-Small for cap-quality classification, and YOLOv8-based water-surface detection for fill-level estimation. It also includes configurable SKU profiles, uncertainty handling, explainability, remediation recommendations, and an operator-facing dashboard.

The topic is industrially relevant because manual quality inspection can be inconsistent, labor-intensive, and difficult to scale on high-throughput production lines. The paper makes a useful effort to move beyond isolated defect classification by proposing an integrated workflow that includes detection, localization, decision logic, remediation, metadata persistence, and operator interaction.

The manuscript provides considerable implementation detail. It describes model architectures, augmentation strategies, focal loss, Monte Carlo Dropout, two-pass cap-region zooming, ONNX and TensorRT deployment plans, fill-ratio equations, training configurations, and edge-deployment considerations.

A particular strength is the two-pass zoom strategy for small cap detection. The paper also transparently reports that the MobileNetV3 classifier and water-surface detector do not meet all predefined targets rather than hiding these limitations. The consolidated performance table makes it relatively easy to compare achieved and targeted metrics.

The paper is relevant to computer vision, intelligent manufacturing, edge AI, and quality decision systems. It demonstrates substantial engineering effort and addresses a practically meaningful problem.

2. Shortcomings in the paper (in detail to be addressed).

The most important concern is the extremely high reported YOLOv11s validation mAP@0.5 of 0.995. Such near-perfect performance may be possible on a small or highly controlled dataset, but the paper does not provide enough information to evaluate whether this result reflects genuine generalization.

The authors do not clearly report:

The total number of images used for each module.

The number of images per class.

The number of unique physical products or bottles.

The training, validation, and test split sizes.

Whether frames from the same video, bottle, batch, or augmented source appear in more than one split.

Whether the reported results are based on a validation set or a completely unseen test set.

The level of class imbalance.

Per-class AP, precision, recall, and confusion matrices.

If adjacent video frames or augmented versions of the same original images were distributed across training and validation sets, the reported metrics may be inflated by leakage. The authors should split the data by production run, physical product instance, acquisition session, or batch rather than by individual image.

The paper repeatedly describes the system as supporting transparent bottles, cans, flexible wrappers, and rigid boxes. However, the experimental results are primarily limited to bottle caps and transparent-bottle water levels. No quantitative results are presented for cans, wrappers, boxes, label defects, packaging tears, contamination, seal defects, corner crush, OCR checks, or QR verification. Therefore, the broad multi-product and multi-defect capability is currently an architectural proposal rather than an experimentally validated system.

The literature-review claims require moderation. Statements such as beverages representing only 4% of food-quality-control studies and no prior system integrating localization, explainability, remediation, and traceability should be tied to clearly verifiable evidence and a reproducible search methodology.

There are also inconsistencies between the broad claims and the reported targets. The system claims image-level PASS/FAIL accuracy of at least 93%, but the paper does not report a final system-level PASS/FAIL metric. It reports separate module metrics instead. A deployable inspection system should be evaluated end to end using product-level false-accept and false-reject rates.

Finally, the paper lacks a publicly available dataset, code repository, annotation protocol, inter-annotator agreement, reproducibility package, and real production-line evaluation.

3. Overall evaluation

Borderline

4. Reviewer's confidence on the overall evaluation of the paper

High

Reviewer #4

Questions

1. Overview and strengths of the paper.

VisionFood QAI presents a compelling, multi-modular deep learning pipeline designed to replace manual visual inspection in high-speed food and beverage lines. The paper excels by directly addressing critical industry gaps, notably the lack of beverage-focused inspection frameworks, edge deployment barriers, and single-defect constraints. Its practical architecture combines lightweight networks—achieving impressive detection metrics like a 0.995 mAP@0.5 for bottle/cap localization with YOLOv11s and 0.915 for fill-level estimation with YOLOv8. The framework stands out by integrating real-world engineering features, including Monte Carlo Dropout for uncertainty quantification, two-pass zoom, and multi-SKU profiles. By bridging deep learning theory with an operator-facing remediation dashboard, the paper delivers strong practical utility.

2. Shortcomings in the paper (in detail to be addressed).

The primary weakness lies in the MobileNetV3-Small cap quality classifier, which achieves only 86.54% accuracy (0.87 F1-score)—an error rate near 13.5% that risks high false rejections on high-speed industrial lines. Furthermore, despite emphasizing edge deployment viability via ONNX and TensorRT, the paper lacks concrete operational latency benchmarks such as processing speed in frames per second (FPS), inference time in milliseconds, or hardware power consumption. The authors should also clarify how uncertain predictions from Monte Carlo Dropout are handled in the inline triage workflow. Lastly, given that dataset scarcity is highlighted as a major research gap, the paper needs greater transparency regarding the scale, class balance, and environmental acquisition conditions (such as conveyor vibration and lighting variation) of the evaluation dataset.

3. Overall evaluation

Borderline

4. Reviewer's confidence on the overall evaluation of the paper

High